BLM3620 Digital Signal Processing

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Lecture #5 – Discrete Time Signals and Convolution

- Basic Discrete Time Signals
- Linear, Time-Invariant, Causal and Stable Systems
- Impulse Response
- Discrete Convolution
- MATLAB Application

Course Materials

Important Materials:

- James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.
- Lizhe Tan, Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Third Edition, Academic Press, 2019.

Auxilary Materials:

- Prof. Sarp Ertürk, *Sayısal İşaret İşleme*, Birsen Yayınevi.
- Prof. Nizamettin Aydin, DSP Lecture Notes.
- J. G. Proakis, D. K. Manolakis, *Digital Signal Processing Fourth Edition*, Peason, 2014.
- J. K. Perin, *Digital Signal Processing*, *Lecture Notes,* Standford University, 2018.

Syllabus

Week	Lectures
1	Introduction to DSP and MATLAB
2	Sinuzoids and Complex Exponentials
3	Spectrum Representation
4	Sampling and Aliasing
5	Discrete Time Signal Properties and Convolution
6	Convolution and FIR Filters
7	Frequency Response of FIR Filters
8	Midterm Exam
9	Discrete Time Fourier Transform and Properties
10	Discrete Fourier Transform and Properties
11	Fast Fourier Transform and Windowing
12	z- Transforms
13	FIR Filter Design and Applications
14	IIR Filter Design and Applications
15	Final Exam

For more details -> Bologna page: http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=5730&aid=3

Remember: Last Lecture

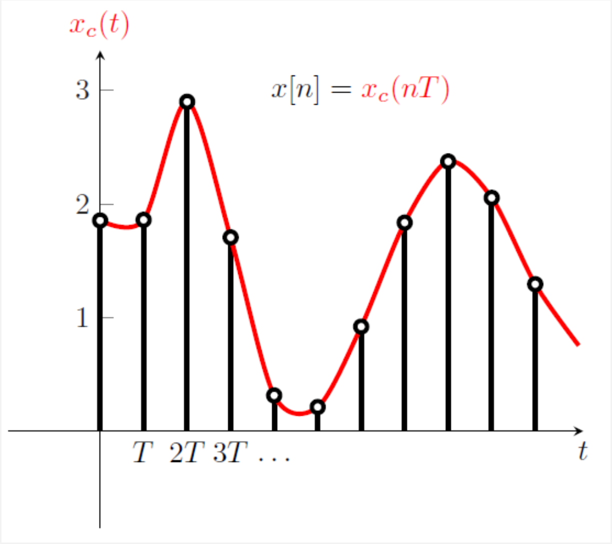
Convolution

Concentrate on the Filtering theory

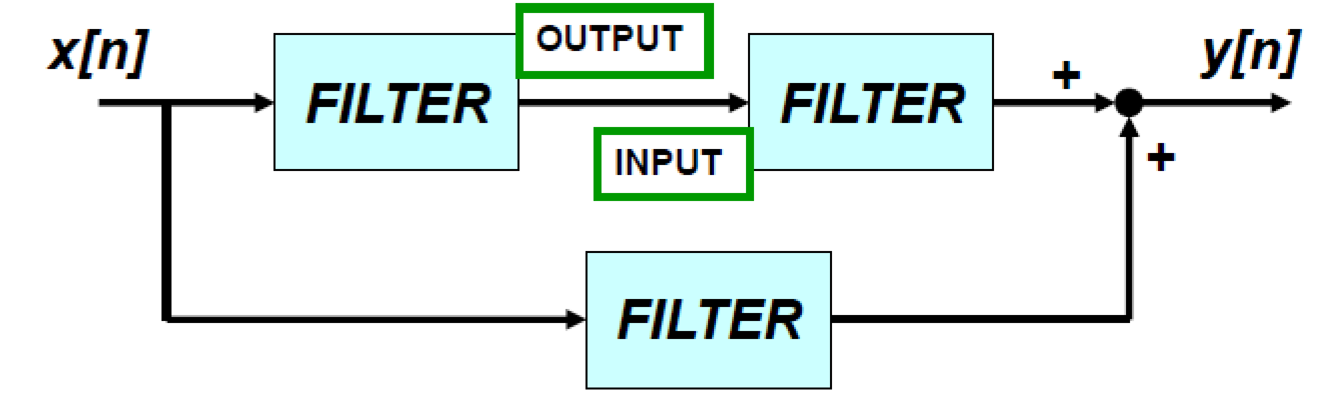
konvolüsyon (Table or Matematik) and grafik

After sampling, we have discrete time signal.

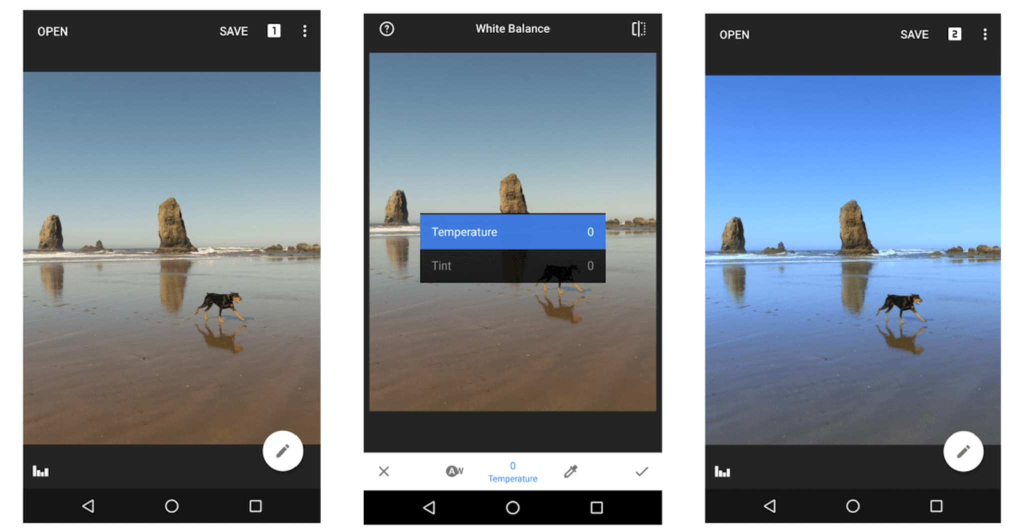
We should learn how to process this digital signal.



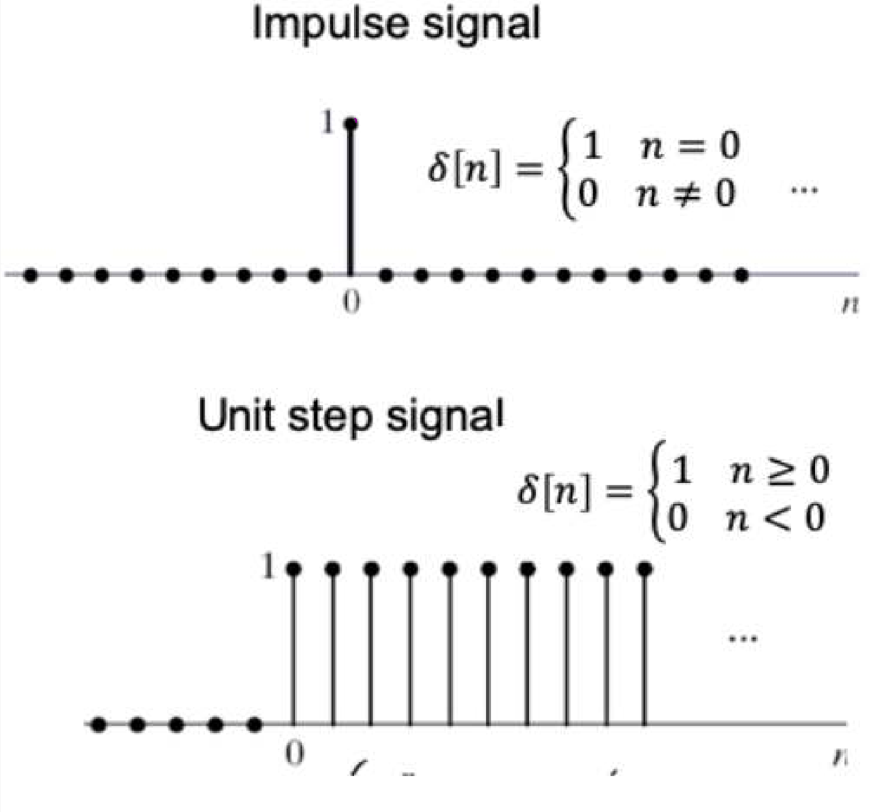
Filtering - Block Diagram Representation

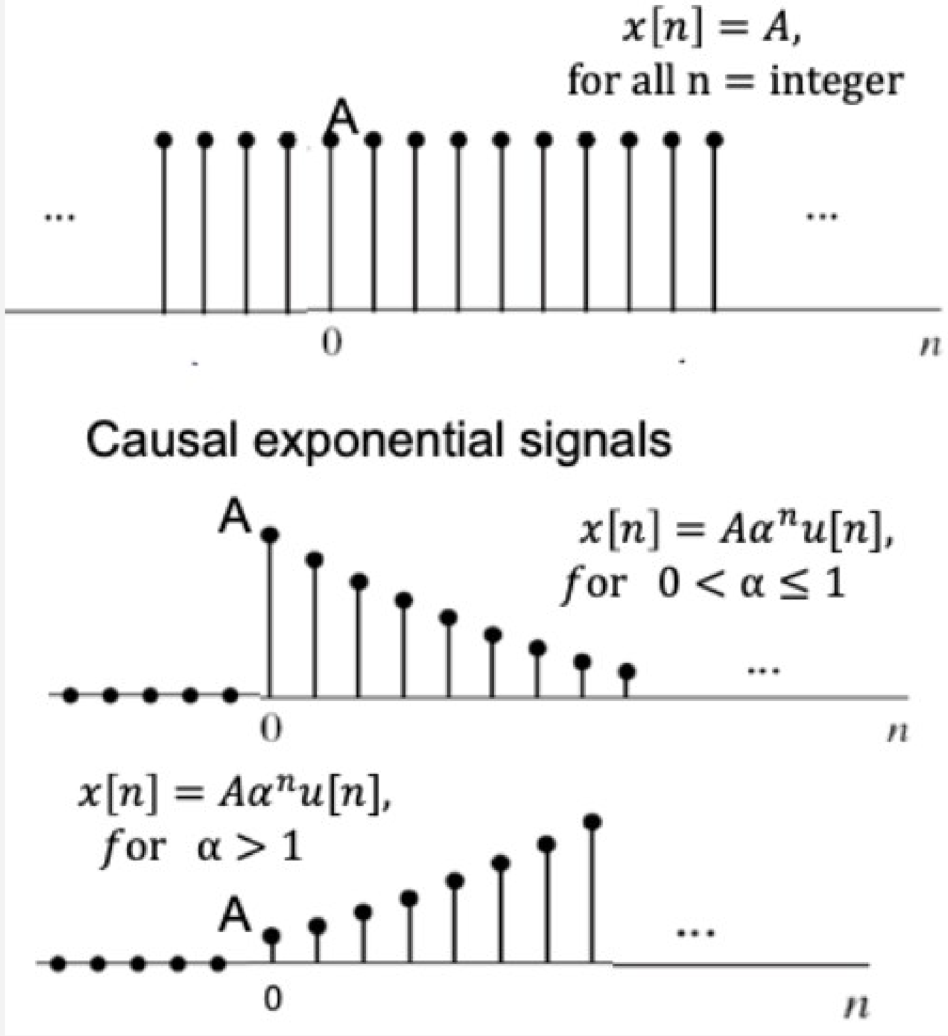


- BUILD UP COMPLICATED FILTERS
  - FROM SIMPLE MODULES
  - Ex: FILTER MODULE MIGHT BE 3-pt FIR

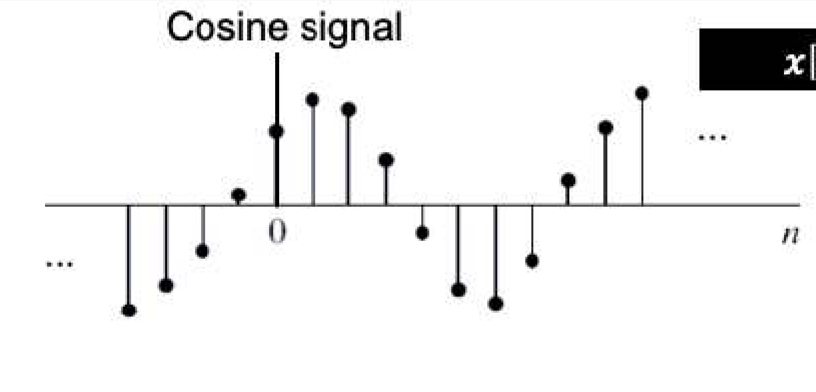


Basic Discrete-Time Signals





Basic Discrete Signals - 2



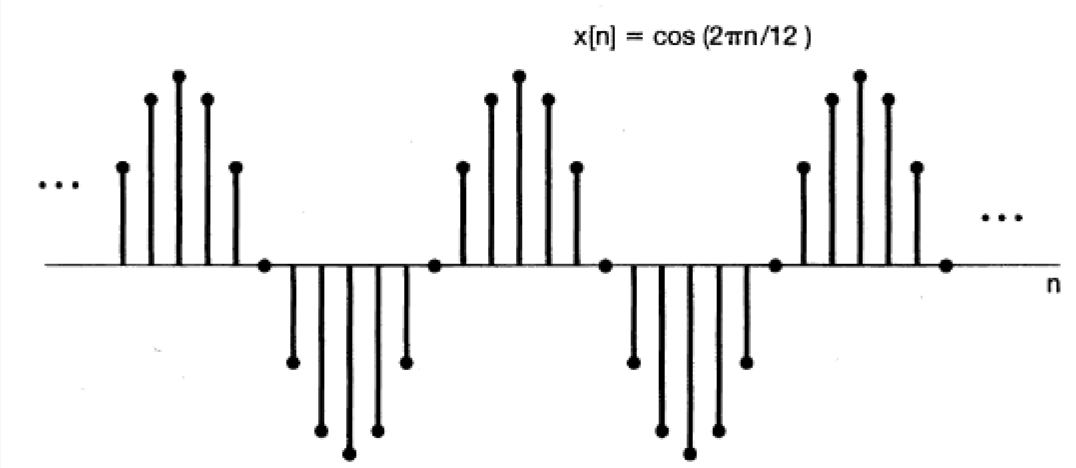
$$x[n] = A\cos(\Omega_0 n + \phi) \quad \text{for n = integer}$$

$\omega = \frac{2\pi f}{f_s}$ $\boxed{\frac{f}{f_s} = \frac{1}{12}}$

$\hat{\omega} = \frac{2\pi f}{f_s} = \frac{2\pi}{12}$

period = 12

$x[n] = \cos\left(\frac{2\pi n}{12}\right)$ -> Plot this signal in MATLAB.



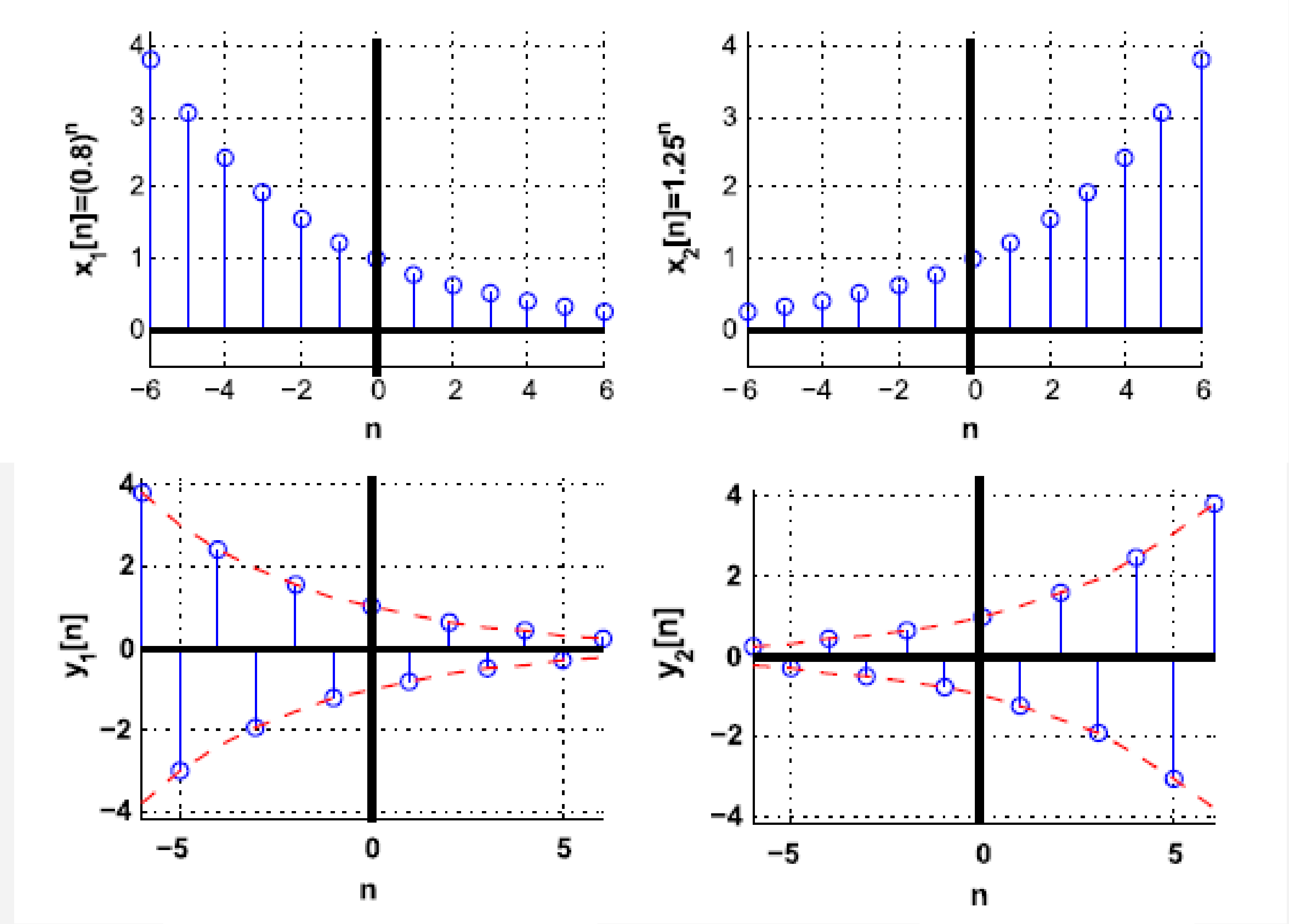
MATLAB code:

```
clc; clear all;
%%

n=-20:20;
x = 0.*n;
wHat = 2*pi/12;
for i=1:length(n)
    x(i)=cos(wHat*n(i));
end

stem(n,x,'filled');
```

Basic Discrete Signals - 3



$y_1[n] = x_1[n]\cos(\pi n)$

$x_2[n]\cos(\pi n)$

MATLAB code:

```
clc; clear all;
%%

n=-20:20;
x = 0.*n;

for i=1:length(n)
   x(i)=(0.8^n(i))*cos(pi*n(i));
end

stem(n,x,'filled');
```

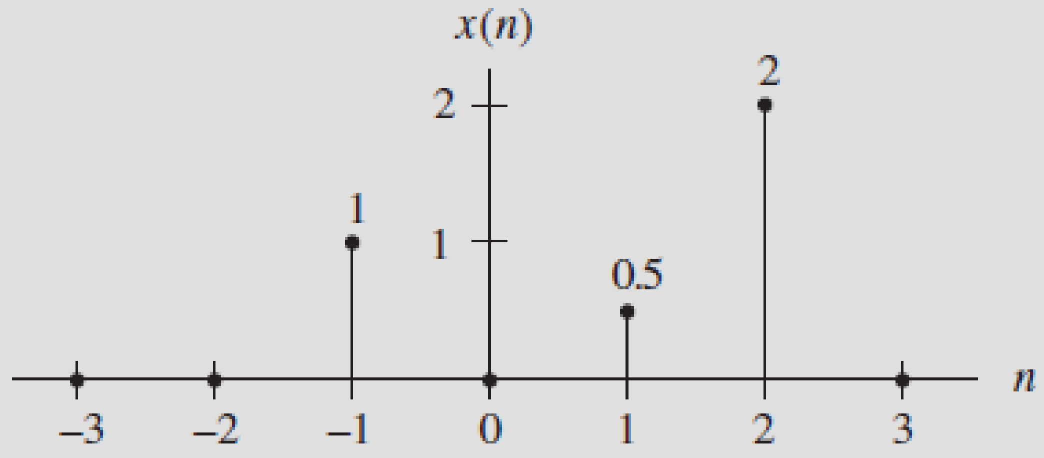
Exponential Sinuzoids

Example

Given the following,

$$x(n) = \delta(n+1) + 0.5\delta(n-1) + 2\delta(n-2),$$

Sketch this sequence.



Example with Sampling

Assuming a DSP system with a sampling time interval of 125 μs,

(a) Convert each of following analog signal $x(t)$ to the digital signal $x(n)$.

1. $x(t) = 10e^{-5000t}u(t)$
2. $x(t) = 10\sin(2000\pi t)u(t)$

(b) Determine and plot the sample values from each obtained digital function.

$t = n.T_s$ $\quad e^{-0.625n}\,u(n)$

$t = n.T_s$ $\quad 10^{-6}.125$

$2000.10^{-6}.125.\pi.n$ $\quad 10\sin(0.25\pi n).u[n]$

(a) Since $T = 0.000125$ s in Eq. (3.3), substituting $t = nT = n \times 0.000125 = 0.000125n$ into the analog signal $x(t)$ expressed in (1) leads to the digital sequence

1. $x(n) = x(nT) = 10e^{-5000\times 0.000125n}u(nT) = 10e^{-0.625n}u(n)$.

   Similarly, the digital sequence for (2) is achieved as follows:

2. $x(n) = x(nT) = 10\sin(2000\pi \times 0.000125n)u(nT) = 10\sin(0.25\pi n)u(n)$

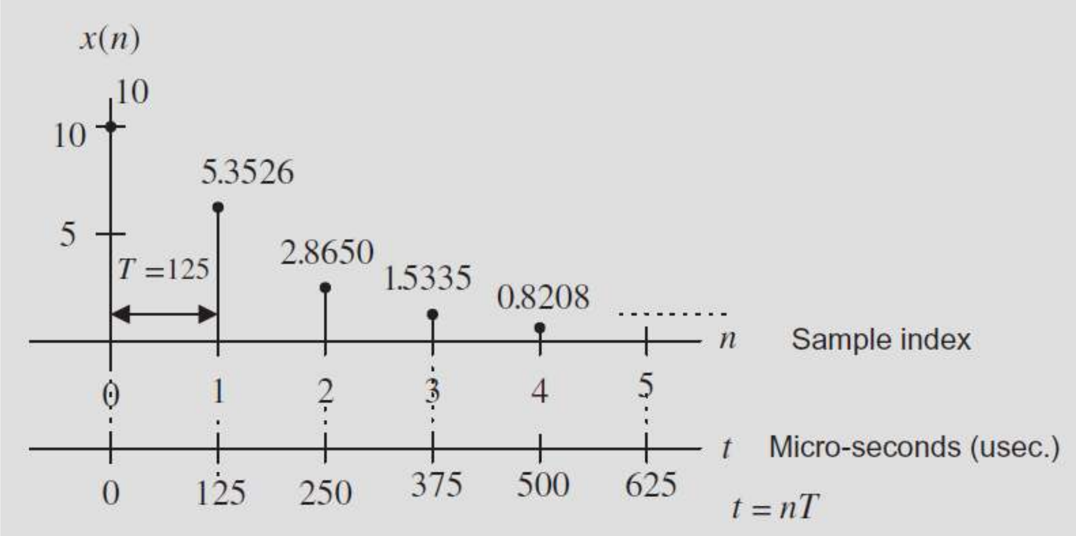
Writing a Sampled Signal in terms of impulse functions

Assuming a DSP system with a sampling time interval of 125 μs,

(a) Convert each of following analog signal $x(t)$ to the digital signal $x(n)$.

1. $x(t) = 10e^{-5000t}u(t)$
2. $x(t) = 10\sin(2000\pi t)u(t)$ -> $10.\sin(2000\ 125 \cdot 10^{-6}\ \pi.n)\ u(n) = 10.\sin(0.250\pi \cdot n)$

(b) Determine and plot the sample values from each obtained digital function.



$$x(0) = 10e^{-0.625\times 0}u(0) = 10.0$$

$$x(1) = 10e^{-0.625\times 1}u(1) = 5.3526$$

$$x(2) = 10e^{-0.625\times 2}u(2) = 2.8650$$

$$x(3) = 10e^{-0.625\times 3}u(3) = 1.5335$$

$$x(4) = 10e^{-0.625\times 4}u(4) = 0.8208$$

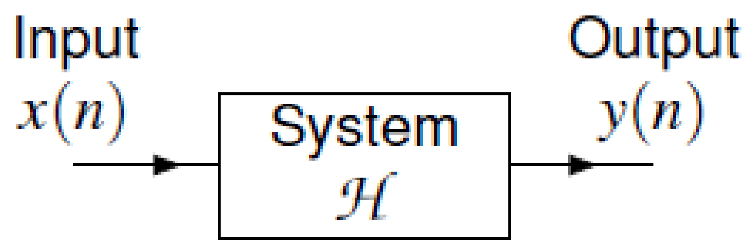
$$x[n] = 10\delta[n] + 5.3526\delta[n-1] + 2.865\delta[n-2] + 1.535\delta[n-3] + 0.821\delta[n-4] + \ldots$$

Discrete Time Systems

- A system with input x and output y can be described by the equation

$$y = \mathcal{H}\{x\},$$

  where $\mathcal{H}$ denotes an operator (i.e., transformation).
- Note that the operator $\mathcal{H}$ ***maps a function to a function*** (not a number to a number).

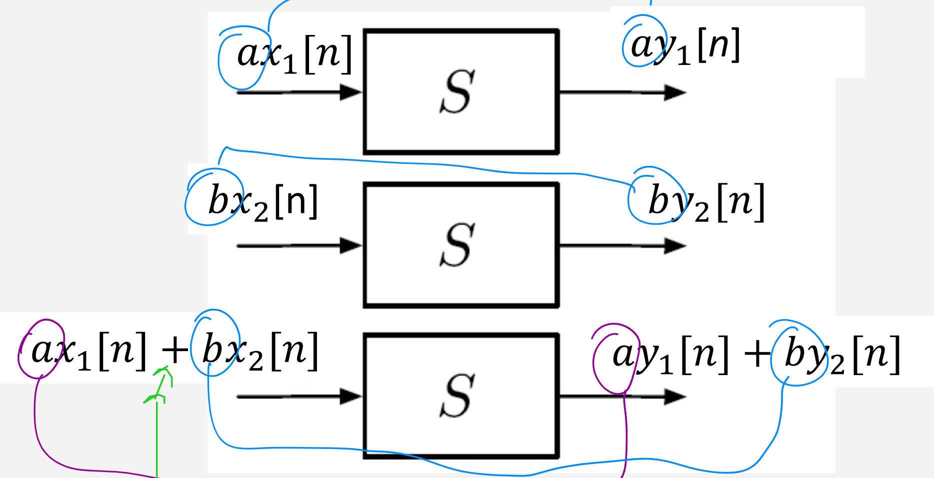


Classification of Systems: (1) Linearity

→ karşılar

A system is linear if and only if it satisfies the superposition principle, or equivalently both the additivity and homogeneity properties.

while push x[n] and if we obtain y[n] → push a.a[n] and we should obtain a.y[n]



bir sistem lineerse ikisini de sağlamalı.

Additivity + Homogeneity Properties

For example, is y(n)=10x(n) system linear? → değer vererek gideceğim

yapsak da sorun yok ama kitabi

$$y_1(n) = 10x_1(n) \text{ and } y_2(n) = 10x_2(n)$$

$$x(n) = \alpha x_1(n) + \beta x_2(n)$$

$$\alpha y_1(n) + \beta y_2(n) = \alpha[10x_1(n)] + \beta[10x_2(n)]$$

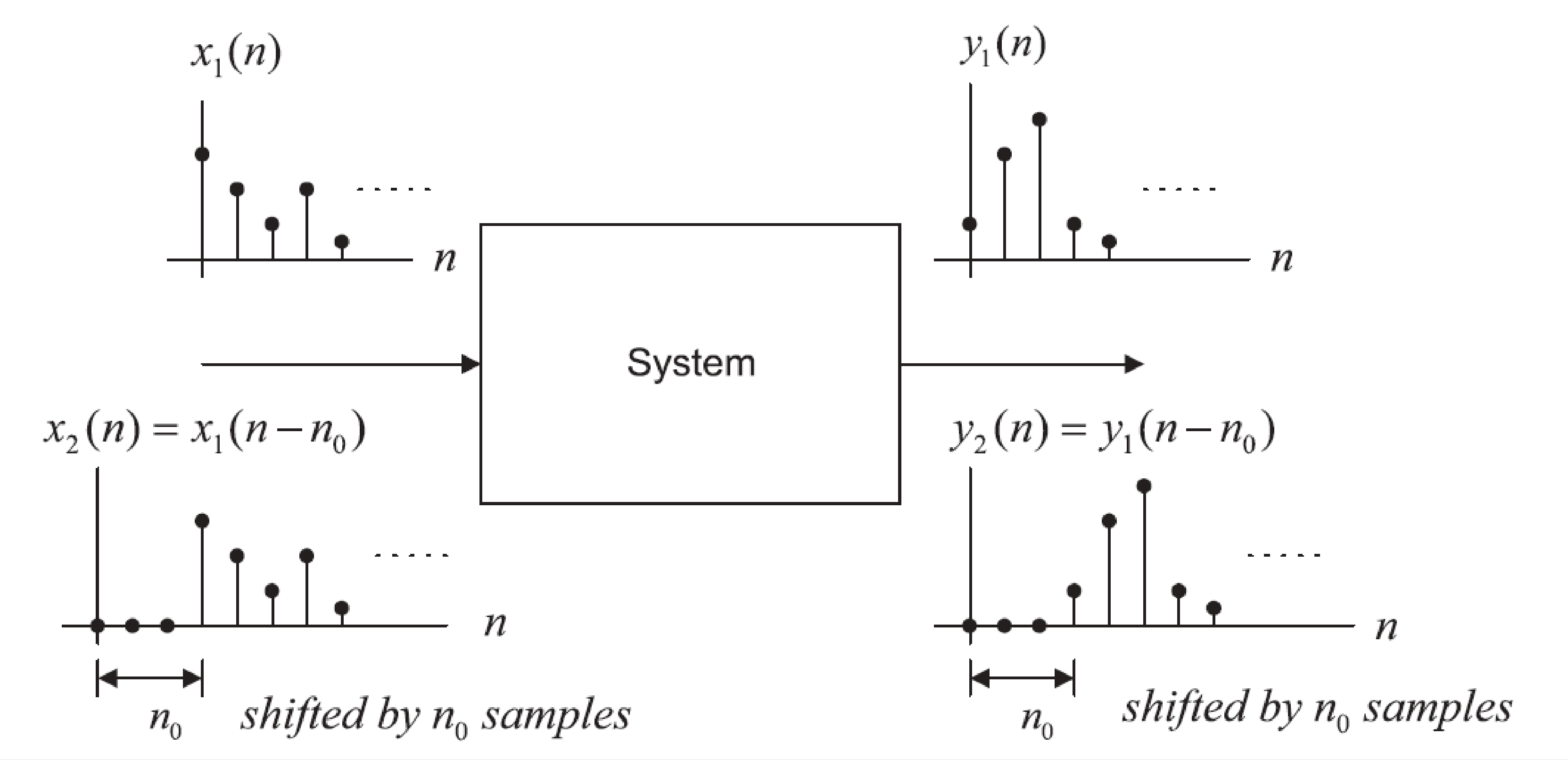
$$y(n) = 10x(n) = 10(\alpha x_1(n) + \beta x_2(n)) = 10\alpha x_1(n) + 10\beta x_2(n)$$

$$y(n) = \alpha y_1(n) + \beta y_2(n)$$

Classification of Systems: (2) Time-Invariance

→ zumanla değişmezlik

If the system is time invariant and $y_1(n)$ is the system output due to the input $x_1(n)$, then the shifted system input $x_1(n_0)$ will produce a shifted system output $y_1(nn_0)$ by the same amount of time n_0.



Example

$y[n] = nx[n]$ -> Is this signal time-invariant or not?

$y[n-1] = [n-1] . x[n-1]$

1) First, apply an input x1 and find the output y1 ->

$$x_1[n] \rightarrow y_1[n] = nx_1[n]$$

2) Second, shift the input x1 by n0 and find the output y2 ->

$[n-1] . x[n-1] \neq x[n-1] . n$

$$x_2[n] = x_1[n-n_0] \longrightarrow y_2[n] = nx_1[n-n_0]$$

$y[n] = x[n] . n$

3) Does it match with the shifted output of y1 ?

y(

$$y_1[n-n_0] = (n-n_0)x_1[n-n_0] \neq y_2[n]$$

$x[n-1] =$ $x[n-1] . [n-1]$

$n . x[n-1]$

Therefore, it is not a time-invariant system!!

Classification of Systems: (3) Causal → nedensel

A causal system is the one in which the output $y(n)$ at time n depends only on the current input $x(n)$ at time n, and its past input sample values such as $x(n-1)$, $x(n-2)$,…. Otherwise, if a system output depends on the future input values such as $x(n+1)$, $x(n+2)$,…, the system is noncausal. The noncausal system cannot be realized in real time.

noncausal
→ $x(n+1)$ → olmaz
n yine da n'den büyük değerler
nedenseldir ör $n-2$, n, $n-1$, $n-5$ . . .

EXAMPLE 3.4

Given the following linear systems

(a) $y(n) = 0.5x(n) + 2.5x(n-2)$, for $n \geq 0$,

(a) Since for $n \geq 0$, the output $y(n)$ depends on the current input $x(n)$ and its past value $x(n-2)$, the system is causal.

→ burası don't care
çünkü çıkış

$$y(n) = 0.25x(n-1) + 0.5x(n+1) - 0.4y(n-1), \text{ for } n \geq 0,$$

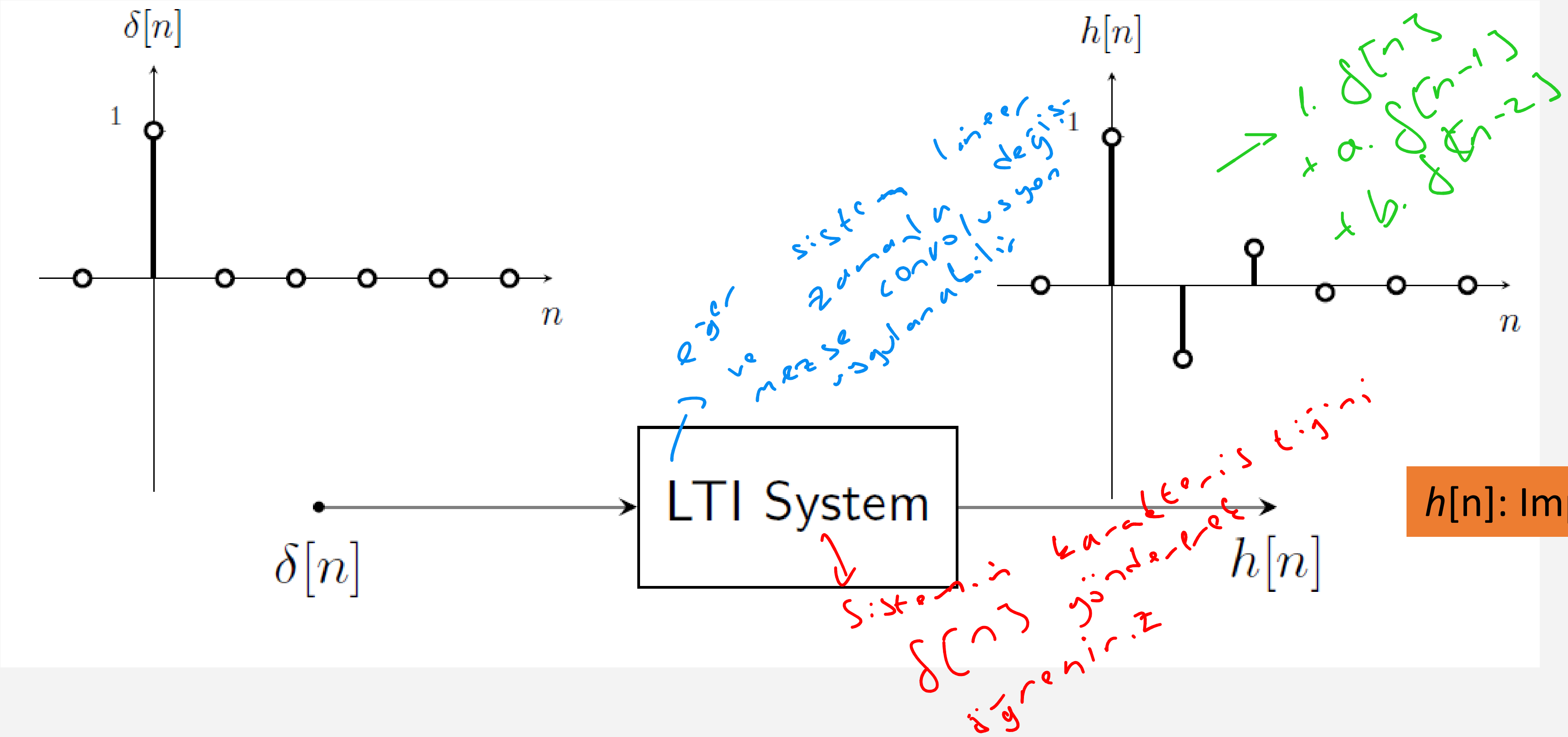
-> Is this causal?

Check This Example at Home!

no yes

System	Linear	Time invariant	Causal
Constant offset $y[n] = x[n] + C, C \neq 0$	N	Y	Y
Time shift $y[n] = x[n - n_d]$	Y	Y	Y, if $n_d > 0$
Squaring $y[n] = x^2[n]$	N	Y	Y
Accumulator $y[n] = \sum_{k=-\infty}^{n} x[k]$	Y	Y	Y
Compressor $y[n] = x[Mn], M > 1$	Y	N	N
Differentiator $y[n] = x[n] - x[n-1]$	Y	Y	Y
A difference equation $y[n] = x[n] + y[n-1]$	Y	Y	Y

Linear and Time-Invariant (LTI) Systems (Very Important!!)



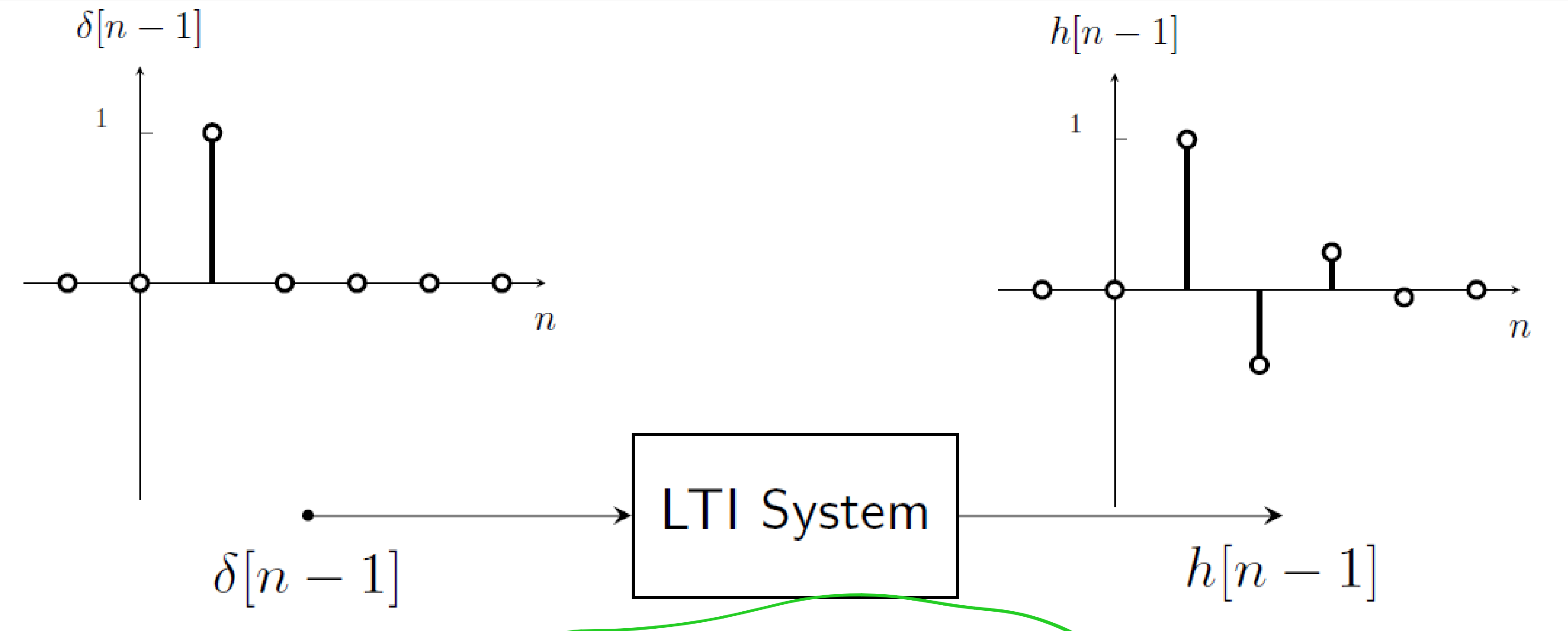
h[n]: Impulse Response

If we write $\delta[n]$ instead of x[n] in system equation, then we can easliy find impulse response h[n]

$y[n] = x[n] + kx[n-d]$ ★ $\Rightarrow$ $\mathrm{h}[n] = \delta[n] + k\delta[n-d]$

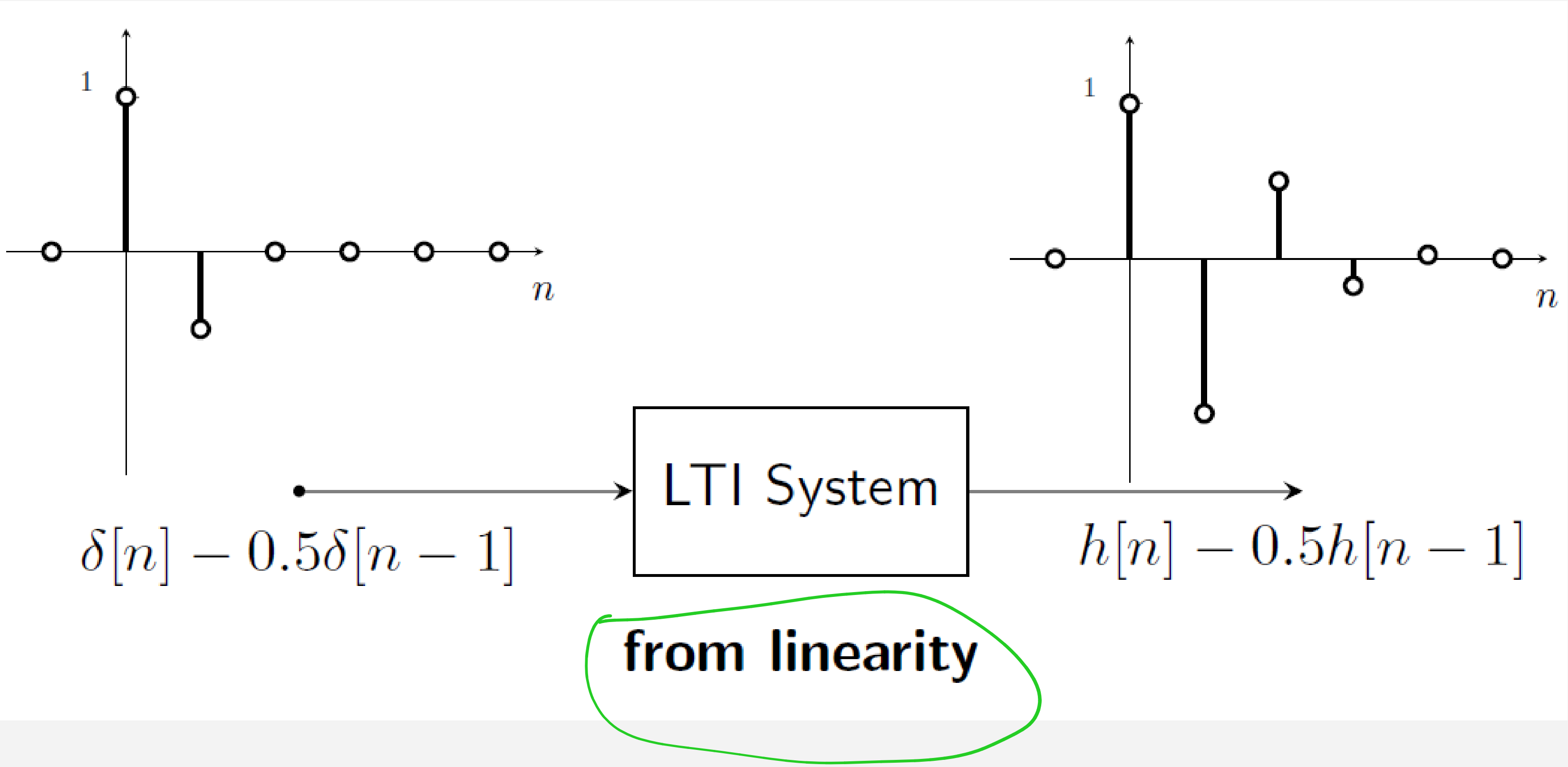
$\delta[n]+k.\delta[n-d]=h[n]$

Linear and Time-Invariant (LTI) Systems (Very Important!!)



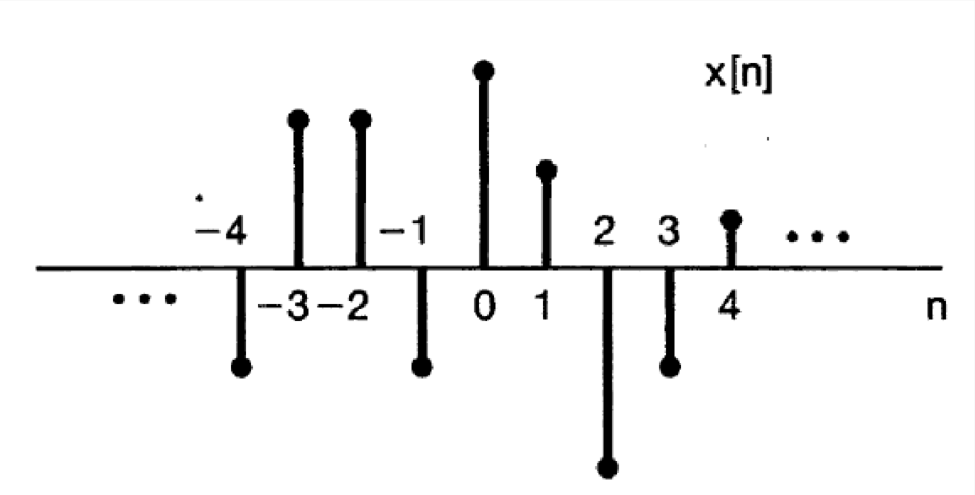
from time invariance

Linear and Time-Invariant (LTI) Systems (Very Important!!)



Remember…

- We can write any sampled signal in terms of impulse functions:



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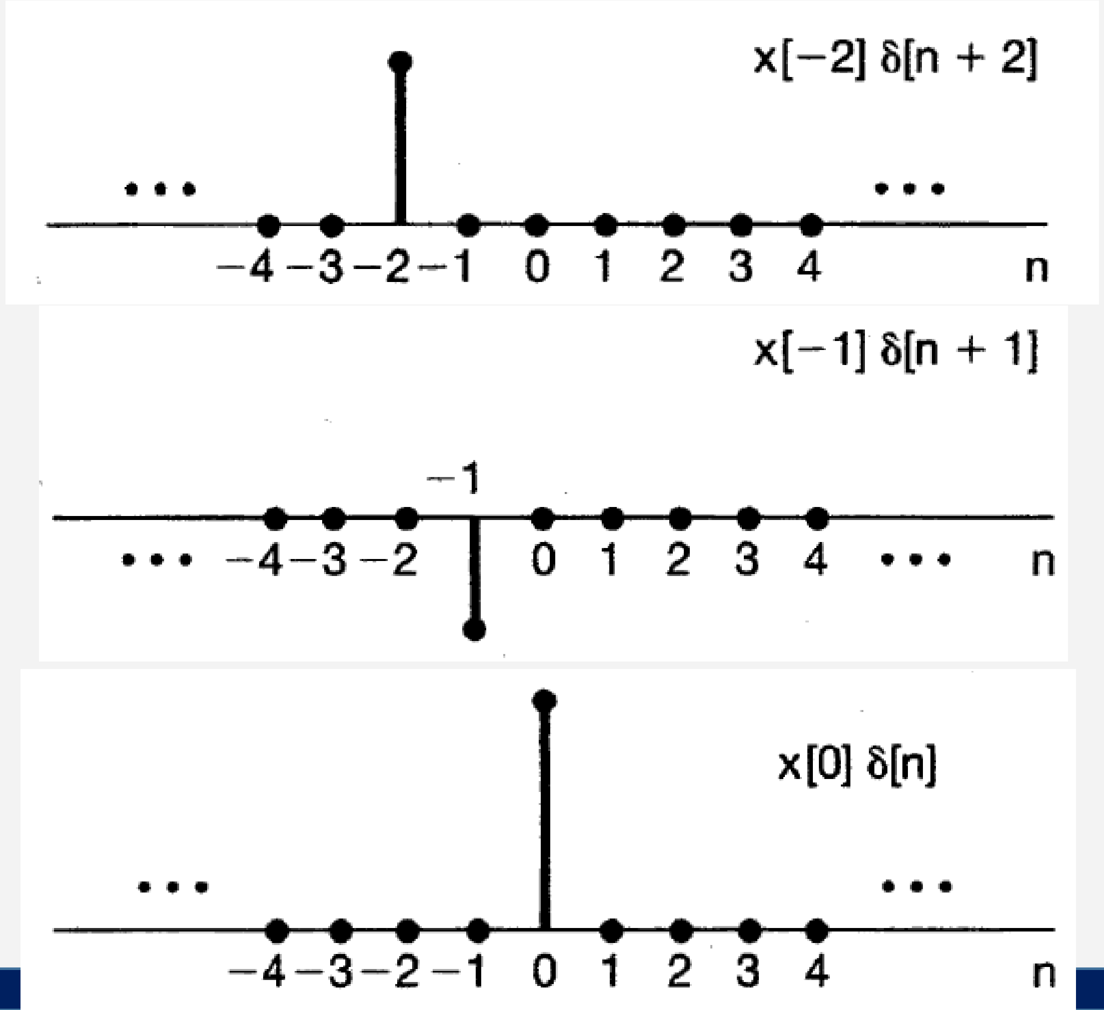


Convolution Sum !!!

$$\sum_{k=-\infty}^{\infty} x[k].h[n-k]$$

$x[n]$ → System → $y[n]$

$$y[n]=\sum_{k=-\infty}^{\infty} x[k].h[n-k]$$

$\delta[n]$ → System → $h[n]$

Or

$$y[n]=\sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$x[k]\delta[n-k]$ → System → $x[k]h[n-k]$

eşit

convolution sum

$$y[n]=x[n]*h[n]$$

$$x[n]=\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$ → System →

$$y[n]=\sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

CONVOLUTION SUM

Properties of LTI Systems

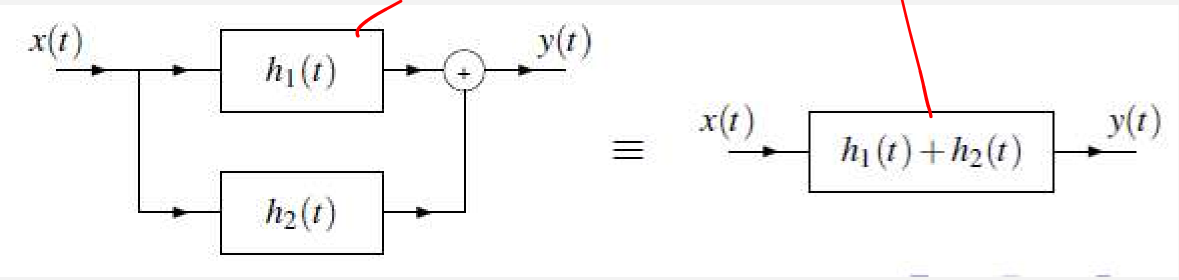
1) Distributive Property:

$$y[n] = x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

bunları ayrı ayrı "*" yapma

toplu sonra "*" yap.

Block Diagram:



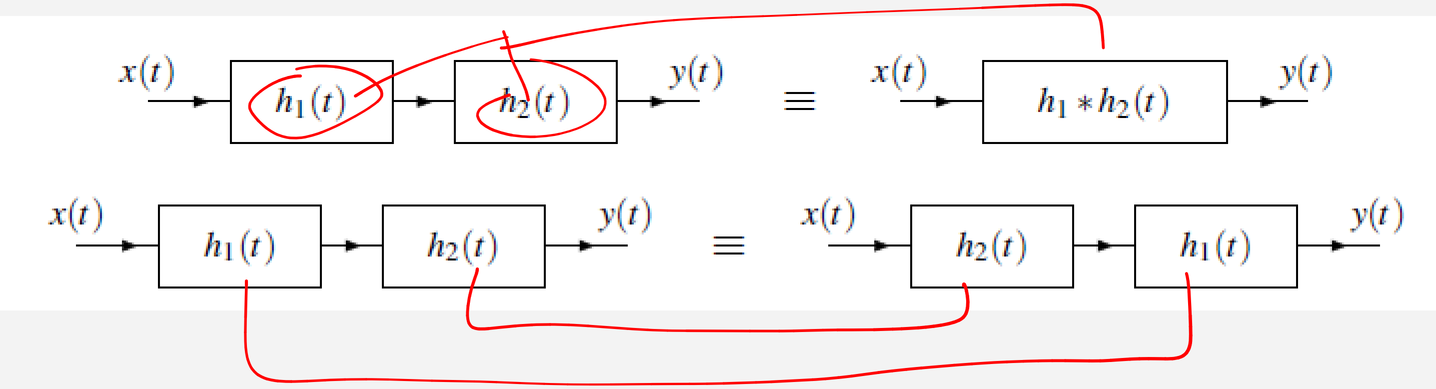
It can be used with the systems connected parallel.

Properties of LTI Systems

2) Associativity Property:

$$y[n] = x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

<u>Block Diyagram :</u>



It can be used with the systems that are cascaded-connected .

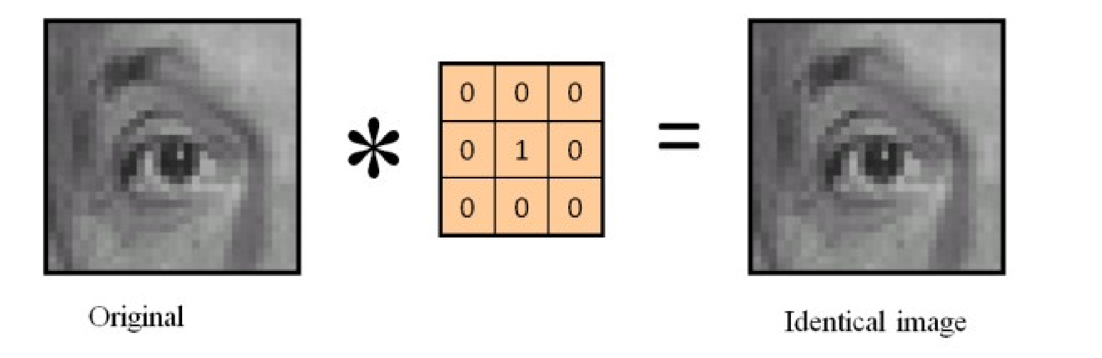
Properties of LTI Systems

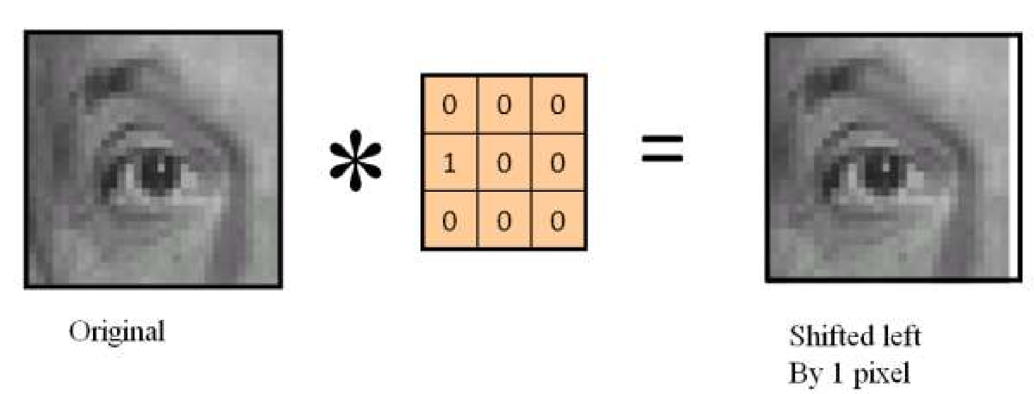
3) Identity Element of Convolution Property:

$$x[n] * \delta[n] = x[n]$$

Convolution with an Impulse

$$x[n] * \delta[n - n_0] = x[n - n_0]$$





Convolution in Discrete-Time Systems

- There are three approaches to calculate convolution:

1) Mathematical Approach

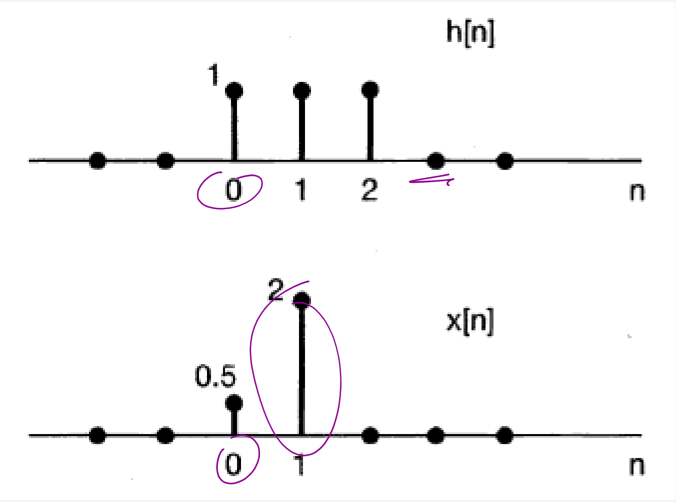
2) Table Approach (Polynomial Multiplication)

3) Graphical Approach

Example 1 (Mathematical Approach)

Impulse response of a LTI system and the input signal are given right.

Find the output y[n]!



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$\sum_{k=0} x[k].h[1-k]$

y[0] = 0.5

y[1] = 0.5 + 2

y[2] = 0.5 + 2

y[3] = 2

n = -1 :

$$y[-1] = \sum_{k=-\infty}^{\infty} x[k]h[-1-k] = \sum_{k=0}^{1} x[k]h[-1-k] = x[0]h[-1] + x[1]h[-2] = 0$$

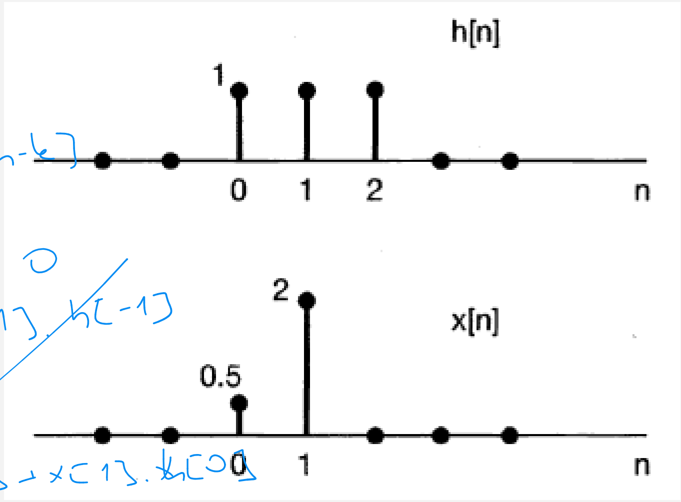
n = 0 :

$$y[0] = \sum_{k=-\infty}^{\infty} x[k]h[0-k] = \sum_{k=0}^{1} x[k]h[-k] = x[0]h[0] + x[1]h[-1] = 0.5$$

Example 1

Impulse response of a LTI system and the input signal are given right.

Find the output y[n]!



$y[0] = \sum_{k=0}^{1} x[k].h[n-k]$

$x[0].h[0] + x[1].\cancel{h[-1]}$ 0

0.5

$y[1] = x[0].h[1] + x[1].h[0]$

$y[2] = x[0].h[2] + x[1].x[1]$

$y[3] = x[0].h[3] + x[1].h[2]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

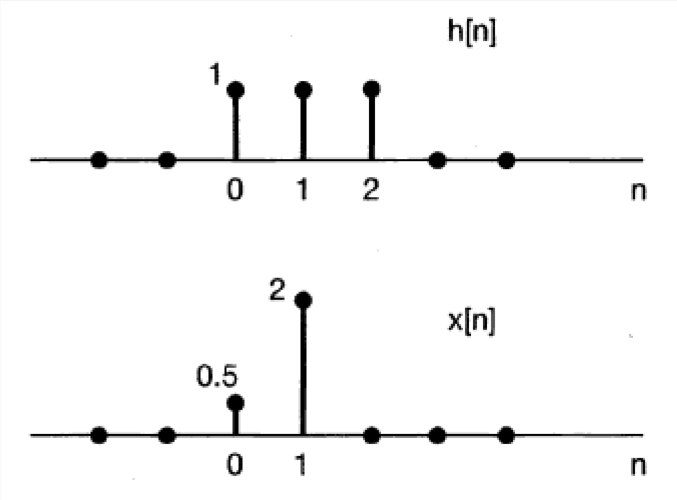
n = 1 :

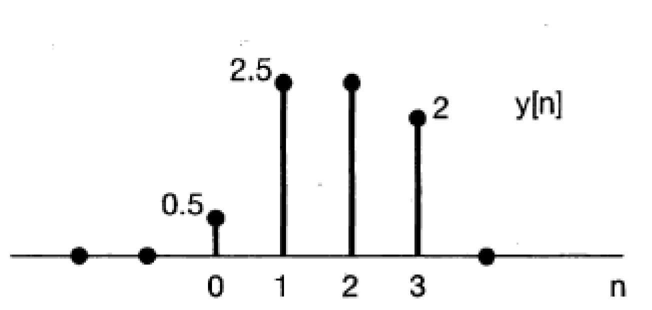
$$y[1] = \sum_{k=-\infty}^{\infty} x[k]h[1-k] = \sum_{k=0}^{1} x[k]h[1-k] = x[0]h[1] + x[1]h[0] = 0.5 * 1 + 2 * 1 = 2.5$$

n = 2 :

$$y[2] = \sum_{k=-\infty}^{\infty} x[k]h[2-k] = \sum_{k=0}^{1} x[k]h[2-k] = x[0]h[2] + x[1]h[1] = 0.5 * 1 + 2 * 1 = 2.5$$

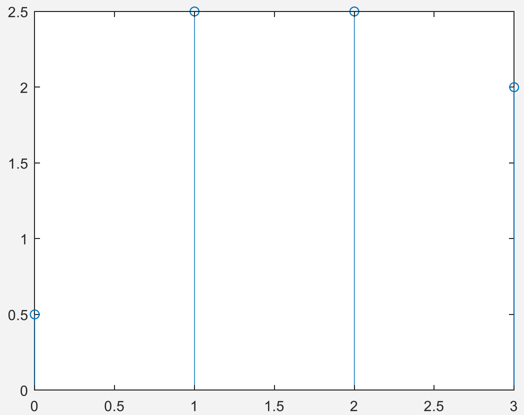
Example 1






```
h = [1 1 1];
x = [0.5 2];

y = conv (x,h);
stem(0:length(h)+length(x)-2,y);
```

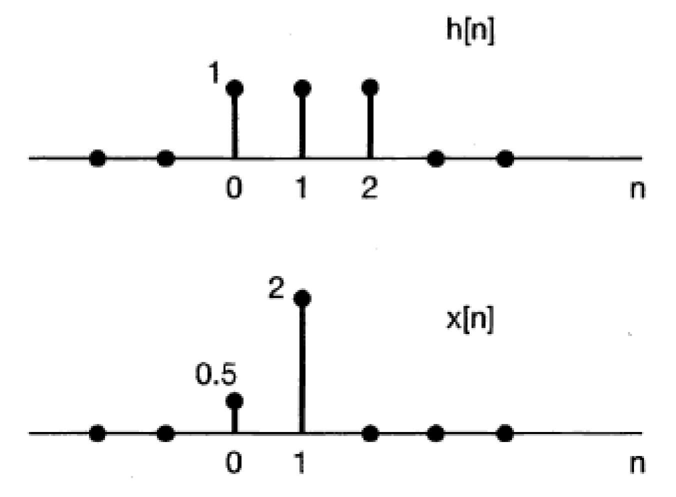


Check the **Associativity property?**

Example 2 (Table Approach)

$$y[n] = \sum_{k=0}^{2} h[k]x[n-k] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2]$$

n	n<0	0	1	2	3	4	5
h[n]	0	1	1	1	0	0	0
x[n]	0	0.5	2	0	0	0	0
h[0]x[n]	0	0.5	2	0	0	0	0
h[1]x[n-1]	0	0	0.5	2	0	0	0
h[2]x[n-2]	0	0	0	0.5	2	0	0
y[n]	0	0.5	2.5	2.5	2	0	0



Graphical approach will be explained next week!!

Another Example of Convolution

Find the impulse response of a system which is given below.

$\delta[n] + \delta[n-1600] \cdot 0.4$

$$y[n] = x[n] + 0.4x[n-1600] \qquad h[n] = \delta[n] + 0.4\delta[n-1600]$$

$\delta[n] + 0.4\,\delta[n-1600]$

$y[n] = x[n] + 0.4 \cdot x[n-1600]$

* Find the output when the input signal $x[n] = \cos(5\pi n)$ is applied to the system.

$\cos(5\pi n) + 0.4 \cdot \cos(5\pi(n-1600))$ $\left(\delta[n] + 0.4 \cdot \delta[n-1600]\right) * \cos(5\pi n)$

$$y[n] = h[n] * x[n]$$

$$y[n] = (\delta[n] + 0.4\delta[n-1600]) * \cos(5\pi n)$$

$\cos(5\pi n) + 0.4 \cdot \cos(5\pi n - 1600)$

***Using Distributive Property**

$$y[n] = \delta[n] * \cos(5\pi n) + 0.4\delta[n-1600] * \cos(5\pi n)$$

***Using Identity Element of Convolution Property**

$$y[n] = \cos(5\pi n) + 0.4\cos(5\pi(n-1600))$$

Convolution with an Impulse

$$x[n] * \delta[n-n_0] = x[n-n_0]$$

Let’s implement this filter to the audio

What does this system do?

$$y[n] = x[n] + 0.4x[n-1600]$$

$$h[n] = \delta[n] + 0.4\delta[n-1600]$$

Write a convolution code that applies this filter to a piano sound.

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

h[n] = [1 0 0 0 0 … 0 0 0 … 1] -> length of h is 1601

$$y[n] = \sum_{k=0}^{1600} h[k]x[n-k]$$

```
clc; clear all;
%%
[x,Fs] = audioread('myRecording.wav');
x = x(:,1); -> sol sağ kanal seçimi
sound(x,Fs);
%%
h = zeros(1,1601);
h(1)=1;
h(1601)=0.8;
%%
lenOutput = length(x)+length(h)-1;
y = zeros(1,lenOutput);

for n = 1:length(x)
    for k = 1:length(h)
        if (n-k)>1
          y(n) =  y(n)+ h(k)*x(n-k);
        end
    end
end
figure (2); stem(1:lenOutput, y));
sound(y,Fs);
```

Just code, no R, L, C circuit elements!!

Convolution in 2D

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Delta fonksiyonu

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Kaydır ve çıkart

$$\begin{bmatrix} -k/8 & -k/8 & -k/8 \\ -k/8 & k+1 & -k/8 \\ -k/8 & -k/8 & -k/8 \end{bmatrix}$$

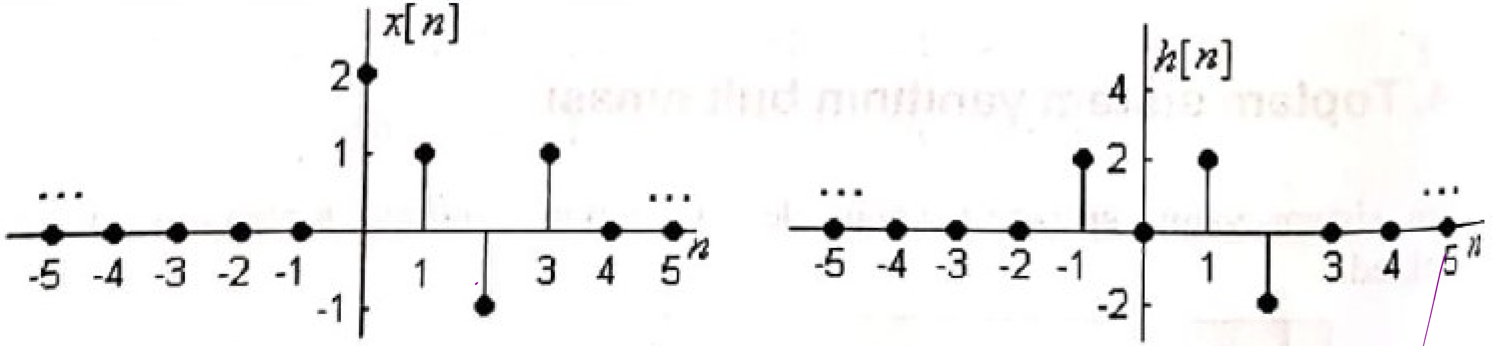
Kenar pekiştirme

/art 2012

MATLAB -> conv2 function

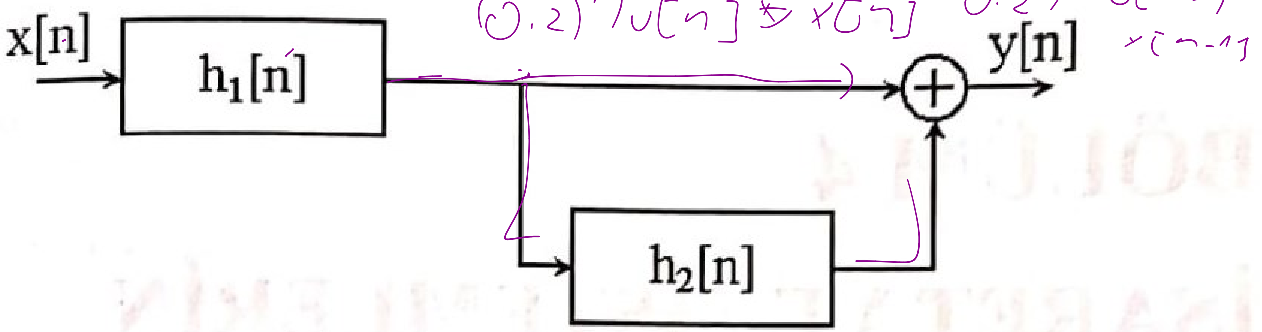
Homework (just for exercise)

3.4. Şekil 3.17'de gösterilen dürtü yanıtı ve giriş işareti için sistemin çıkışını bulunuz. Bu sistem kararlı mıdır? Bu sistem nedensel midir?



Şekil 3.17. Problem 3.4. için giriş işareti ve dürtü yanıtı.

3.5. Şekil 3.18'de gösterilen sistem düzeneğinde $h_1[n] = (0.2)^n u[n]$ ve $h_2[n] = \delta[n-1]$ olarak verildiğine göre eşdeğer sistemin dürtü yanıtını bulunuz.



Şekil 3.18. Problem 3.5. için sistem düzeneği